1. Write the function $f(x) = x^2 + 1$ on the domain {0,1,...,5} as a set.

Let $I = \{0, 1, ..., 5\}$

Let $O = \{f(0), f(1), ..., f(5)\}$

$O = \{1, 2, 5, 10, 17, 26\}$

2. What is the difference between the range and the image of a function? What can we say about functions where the two are equal?

- The range of a function is the set of all possible outputs that could be met (e.g. the range of $f(x) = x^2$ would be $\mathbb{R}^+$)
- The image of a function is the set of all outputs that are 'hit' when inputs are passed in.

3. Consider the function $f : \{0, 1, ..., 9\} \to \{a, b, ..., z\}$ that maps digits to the first letter of the digit, when spelled out as an English word. For example, $f(0) = z$ because 0 is spelled 'zero'.

   a. What is the image of f?

   $\text{im}(f) = \{z, o, t, f, s, e, n\}$

   b. What is the preimage of the letter 's' under f?

   $\text{im}^{-1}(f, s) = \{6, 7\}$ *(funny)*

   c. What is the (direct) image of $\{1, 2, 3\}$ under f?

   $\text{im}(f, \{1, 2, 3\}) = \{o, t\}$

4. For $A = \{0, 1, 2\}$, compute the set $\{(3x + 1) \bmod 6 \mid x \in A\}$, where $a \bmod b$ is the remainder when dividing a by b. How many elements does this set have?

Applying the operation to A gives: $\{4, 1\}$, giving 2 elements.

5. Give an example of a surjective function that does not have an inverse.

$f(x) = x^2$. This is because f is not a 1:1 function, so every output could have more than one input.

6. Consider the function $f : \{0, 1, ..., 9\} \to \{a, b, ..., z\}$ that maps digits to the first letter of the digit, when spelled out as an English word. Is f injective, surjective, both (bijective), or neither?

Neither. It's not injective, as some inputs have the same outputs (e.g. $f(6) = f(7)$). It's not surjective either, as not all outputs are mapped with an input (e.g. a is not mapped to any inputs). Since it's neither injective nor surjective, it's not bijective either.

7. For each of the following functions, determine whether they're injective, surjective, or bijective.

   a. $f(x) = x + 1, x \in \mathbb{N}$

   Injective ✓: All inputs have a unique output.

   Surjective ✗: 0 does not have a natural input.

   Bijective ✗: Injective but not surjective.

b. $f(x) = x^2, x \in \mathbb{N}$

Injective ✓: All inputs map to a unique output.

Surjective ✗: Not all outputs map to at least one input.

Bijective ✗: Injective but not surjective.

c. $f(x) = x^2, x \in \mathbb{R}$

Injective ✗: Not all inputs map to unique outputs.

Surjective ✗: Not all outputs map to at least one input.

Bijective ✗: Neither injective nor surjective.

d. $f(x) = e^x, x \in \mathbb{R}$

Injective ✓: All inputs map to a unique output.

Surjective ✗: Not all outputs map to at least one input.

Bijective ✗: Injective, but not surjective.

e. $f(x) = e^x, f : \mathbb{R} \to \mathbb{R}^+$

Injective ✓: All inputs map to a unique output.

Surjective ✓: All outputs map to at least one input.

Bijective ✓: Both injective and surjective.

f. Logical negation ($\neg$) as a function $V \to V$, where $V = \{T, F\}$

Injective ✓: All inputs map to a unique output.

Surjective ✓: All outputs map to at least one input.

Bijective ✓: Both injective and surjective.

8. Find a non-trivial bijection between $(0, 1)$ and $(1, \infty)$. What is the inverse of your function?

$f(x) = \frac{1}{x}$

$f^{-1}(x) = \frac{1}{x}$

9. On the set $A = \{0, 1, 2\}$, consider $f(a) = a + 1 \bmod 3$ and $g(a) = 2a \bmod 3$. What is $g \circ f$ as a set and as a formula?

$g \circ f \equiv g(f(a)) = 2(a + 1 \bmod 3) \bmod 3$

$g \circ f = \{2, 1, 0\}$

10. Note that not all books define function composition the same way. Use the definition in the lecture.

    a. On $\mathbb{R} \to \mathbb{R}$, consider the functions $f(x) = 2x$, $g(x) = x + 1$, and $h(x) = x^2$. What is the function $k = f \circ h \circ g$, both as a function and evaluated at $x = 4$?

    $f \circ h \circ g = f(h(g(x)))$

    $k = f(h(x + 1))$

$= f((x+1)^2)$

$= 2((x+1)^2)$

$= 2(x^2 + 2x + 1)$

$= 2x^2 + 4x + 2$

$f \circ h \circ g$ at $4 = 50$

b. The same as above, but for the function $j = g \circ h \circ f$?

$g \circ h \circ f = g(h(f))$

$j = g(h(2x))$

$= g((2x)^2)$

$= (2x)^2 + 1$

$= 4x^2 + 1$

c. On the set $S = \{1, 2, 3, 4\}$, consider the functions f, g with signature $S \to S$ defined as $f = \{(1, 2), (2, 4), (4, 1), (3, 3)\}$ and $g = \{(1, 3), (3, 1), (2, 2), (4, 2)\}$.
What is $f \circ g \circ g$ both as a function, and evaluated at 2?

$f \circ g \circ g = f(g(g))$

$f \circ g \circ g$

$| 1 \to 2$

$| 2 \to 4$

$| 3 \to 3$

$| 4 \to 4$

11. $f : A \to B$ and $g : B \to C$. If $g \circ f$ is injective, what can you say about f and g?

f is injective, g doesn't have to be injective. This is because f needs to have unique outputs, for g to produce unique outputs. g doesn't necessarily need to be injective, as it only needs to map unique outputs to the image of f, not the entire set of B.